SAROSH KUMAR

+91-7619562240 | saroshkumar00@gmail.com | Hubli, Karnataka, India
LinkedIn | GitHub | Portfolio

Summary

Computer Science undergraduate (2024–2028) with hands-on experience in scientific machine learning, deep learning, audio AI, and low-latency backend systems. Proven track record in training and deploying neural architectures, streaming inference pipelines, and physics-informed discovery engines. Active open-source contributor across research organizations (BrainGlobe, ML-LAM, ML4SCI).

Education

KLE Technological University, Hubli 2024 – 2028
B.E. in Computer Science & Engineering **CGPA: 8.06 / 10**

Technical Skills

- **Languages:** Python, C++, Java, C, SQL, JavaScript, HTML/CSS
- **Deep Learning & Audio AI:** PyTorch, Torchaudio, Transformers, WavLM, LCNN, LFCC, librosa, Faster-Whisper, PyG (GNNs)
- **Machine Learning & Data:** Scikit-Learn, XGBoost, Prophet, NumPy, Pandas, SciPy, Streamlit, Plotly, Parquet
- **Systems & Tools:** FastAPI, WebSockets, RESTful APIs, Docker, Linux, Git, GitHub, Groq API, PyAutoGUI
- **Core CS Fundamentals:** Data Structures & Algorithms, Operating Systems, DBMS, Concurrency, System Design

Open Source Contributions

- **BrainGlobe Initiative (`cellfinder`):** Enhanced production 3D image processing and classification pipelines in Python/PyTorch, optimizing tensor operations and multi-dimensional array evaluation for computational neuroscience workflows.
- **ML-LAM Collaboration (`neural-lam`):** Contributed to graph neural weather prediction framework, working with Graph Neural Network (GNN) architectures and training utilities in PyTorch.
- **ML4SCI (`DeepLense`):** Modernized MPS device detection for Apple Silicon acceleration, resolved micro-AUROC metric issues in transformer classification pipelines, and cleaned deprecated scheduler parameters.

Projects

Voice Integrity — Real-Time Deepfake Voice Detection Engine *PyTorch, WavLM, LFCC-LCNN, FastAPI*
[Live Demo] | [Hugging Face Models] | [GitHub]

- Open-sourced anti-spoofing models on Hugging Face (LFCC-LCNN & WavLM Base+), engineering a 7,000 Hz low-pass parity gate to eliminate resampler spectral bias.
- Ensembled spectral and transformer logits via Platt scaling and EMA smoothing, achieving 91.5% zero-day recall at 1.7% false alarm rate across 25 unseen generators; built sub-250 ms streaming FastAPI WebSocket pipeline with RMS VAD.

ExoTerra — NASA Exoplanet Characterization & Discovery Engine *Python, Scikit-Learn, NASA TAP API, Streamlit*
[GitHub Repository]

- Ingested and cached 5,800+ confirmed exoplanets via NASA's TAP API with Chen & Kipping empirical mass-radius relations and Stefan-Boltzmann flux modeling; implemented Kopparapu Habitable Zone polynomial boundaries and Earth Similarity Index (ESI).
- Deployed an unsupervised ML suite: Silhouette-optimized K-Means ($k = 4$), GMM boundary transition entropy, 3D PCA capturing **89.2% cosmological variance**, and Isolation Forest detecting extreme celestial anomalies.

Air Quality Forecasting & Source Attribution Engine *Python, XGBoost, Prophet, FastAPI, OpenAQ, Groq*

- Engineered an autoregressive 24-hour AQI forecasting pipeline using recursive XGBoost on 3-year OpenAQ data, benchmarked against Prophet and persistence baselines.
- Developed a heuristic attribution engine computing emission breakdown (Traffic, Industry, Construction) using PM10/PM2.5 ratios, chemical markers (NO_2, SO_2), and wind vectors with dynamic station retraining endpoints in FastAPI.

Leadership & Experience

Lead — Data Science Club, KLE Technological University Hubli, India

- Spearheaded hands-on learning initiatives, technical workshops, and masterclasses in ML/data science, mentoring a community of 500+ students.
- Directed the planning and execution of club hackathons and tech talks, boosting cross-batch peer learning and engagement in emerging technologies.

Achievements & Hackathons

- **Top Hackathon Finalist:** Shortlisted among top finalist teams across premier hackathons including **HackTU 7.0** (Thapar Institute, MLH; top finalists out of 6,000+ teams), **Hack-Nocturne 2.0** (SMVIT Bengaluru), and **Anvil** (Scaler Bengaluru).
- **HACK4SOC 3.0 (RVCE Bengaluru): 4th Place** finalist in state-level inter-college hackathon.
- **IGNITRIX Agentic AI Hackathon:** Awarded Consolation Prize for autonomous agent system design.
- **Word Whizzle (KLE Tech):** Secured 3rd Place.